Module 7: Signing and Cryptography

Ed25519 Digital Signatures & Key Management

Cryptographic Foundations

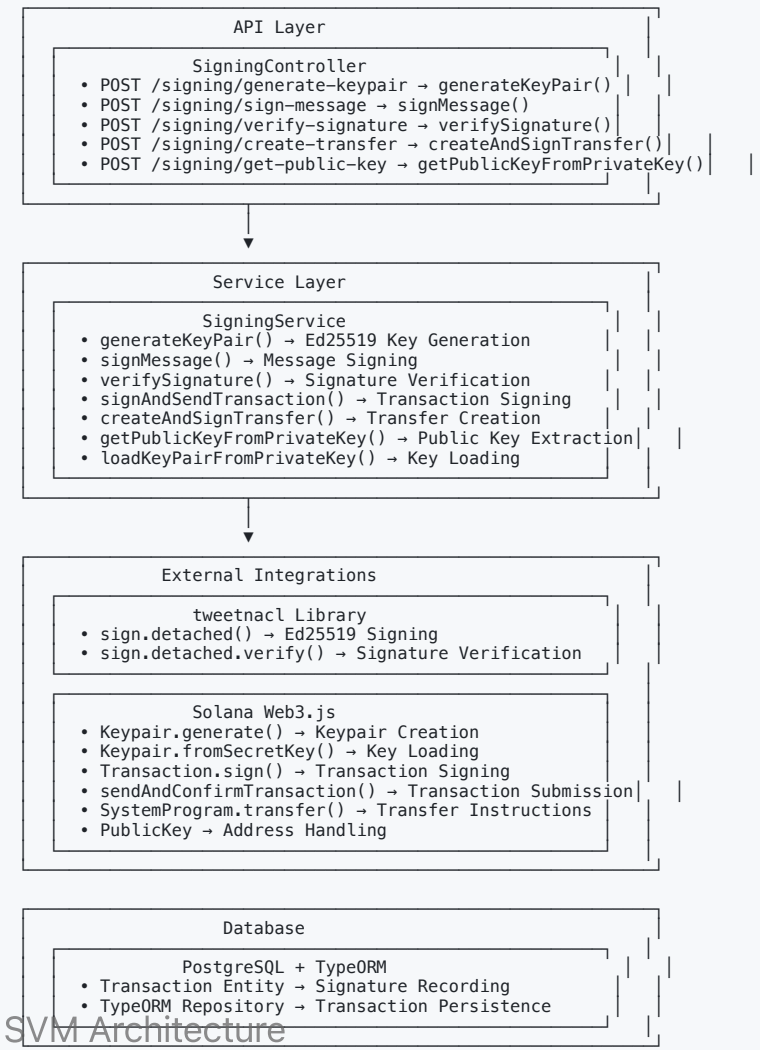
Ed25519 Digital Signatures

- **Algorithm**: Ed25519 elliptic curve cryptography
- **Key Size**: 256-bit keys (32 bytes)
- **Signature Size**: 512-bit signatures (64 bytes)
- **Security**: Post-quantum resistant design
- **Performance**: Fast signing and verification

Solana's Crypto Requirements

- **Transaction Signing**: All transactions must be signed
- **Account Security**: Private keys control account ownership
- **Program Security**: Signed deployments and upgrades
- **Multi-signature**: Multiple key validation

Architecture Overview



Key Management

Key Generation

```
generateKeyPair(): KeyPairResponse {
  // Generate new Ed25519 keypair
  const keypair = Keypair.generate();

  // Return only public key (security best practice)
  return {
    publicKey: keypair.publicKey.toString(),
    // private key is NEVER returned via API
  };
}
```

Private Key Handling

```
loadKeyPairFromPrivateKey(privateKeyBase64: string): Keypair {
  try {
    // Decode base64 private key
```

Message Signing

Signing Workflow

```
async signMessage(signMessageDto: SignMessageDto): Promise<SigningResult> {
  try {
    // Load keypair from private key
    const keypair = this.loadKeyPairFromPrivateKey(signMessageDto.privateKey);

    // Decode message from base64
    const messageBytes = Buffer.from(signMessageDto.message, 'base64');

    // Sign message using Ed25519
    const signature = nacl.sign.detached(messageBytes, keypair.secretKey);

    // Return base64 encoded signature
    return {
      signature: Buffer.from(signature).toString('base64'),
      publicKey: keypair.publicKey.toString(),
      success: true
    };
  } catch (error) {
    throw new BadRequestException('Message signing failed');
```

Signature Verification

Verification Process

```
async verifySignature(verifyDto: VerifySignatureDto): Promise<VerificationResult> {
  try {
    // Decode inputs from base64
    const messageBytes = Buffer.from(verifyDto.message, 'base64');
    const signatureBytes = Buffer.from(verifyDto.signature, 'base64');
    const publicKey = new PublicKey(verifyDto.publicKey);

    // Verify signature using Ed25519
    const isValid = nacl.sign.detached.verify(
      messageBytes,
      signatureBytes,
      publicKey.toBytes()
    );

    return {
      isValid,
      publicKey: publicKey.toString(),
      message: isValid ? 'Signature is valid' : 'Signature is invalid'
    };
  } catch (error) {
    throw new BadRequestException('Signature verification failed');
```

Transaction Signing

Complete Transaction Flow

```
async createAndSignTransfer(transferDto: CreateTransferDto): Promise<TransactionResult> {
  try {
    // Load sender's keypair
    const senderKeypair = this.loadKeyPairFromPrivateKey(transferDto.privateKey);

    // Create transfer instruction
    const transferInstruction = SystemProgram.transfer({
      fromPubkey: senderKeypair.publicKey,
      toPubkey: new PublicKey(transferDto.toAddress),
      lamports: transferDto.amount
    });

    // Build transaction
    const transaction = new Transaction().add(transferInstruction);

    // Get recent blockhash
    const { blockhash } = await this.connection.getRecentBlockhash();
    transaction.recentBlockhash = blockhash;
    transaction.feePayer = senderKeypair.publicKey;

    // Sign transaction
    transaction.sign(senderKeypair);

    // Submit to network
    const signature = await this.connection.sendAndConfirmTransaction(transaction);

    // Record transaction in database
    await this.recordTransaction(signature, transferDto);

    return {
      signature,
      success: true,
      transactionId: signature
    };
  } catch (error) {
```

Transaction Workflow

Signing Workflow Steps

```
1. 🔑 Load Keypair → Private Key to Keypair
2. 🏗️ Build Transaction → Instruction Assembly
3. ✍️ Sign Transaction → Ed25519 Signing
4. 📡 Submit to Network → sendAndConfirmTransaction
5. 💾 Record Transaction → Database Persistence
```

Transaction Components

- **Instructions**: Program calls and data
- **Recent Blockhash**: Prevents replay attacks
- **Fee Payer**: Account paying transaction fees
- **Signatures**: Ed25519 signatures from required signers

API Endpoints

Key Management

- `POST /signing/generate-keypair` - Generate new Ed25519 keypair
- `POST /signing/get-public-key` - Extract public key from private key

Message Signing

- `POST /signing/sign-message` - Sign arbitrary message
- `POST /signing/verify-signature` - Verify message signature

Transaction Operations

- `POST /signing/create-transfer` - Create and sign SOL transfer

Response Formats

```
interface KeyPairResponse {
```

Security Considerations

Implemented Security Measures

- **Private Key Protection**: Never returned via API responses
- **Input Validation**: Strict format checking for all inputs
- **Error Sanitization**: No sensitive data in error messages
- **Base64 Encoding**: Safe transport of binary cryptographic data
- **Transaction Recording**: Complete audit trail

Cryptographic Best Practices

- **Ed25519 Standard**: Industry-standard elliptic curve cryptography
- **Deterministic Signatures**: Consistent signature generation
- **Replay Protection**: Blockhash prevents transaction replay
- **Multi-signature Ready**: Supports multiple signer validation

MPC Integration (Future)

Multi-Party Computation Features

- **Key Share Distribution**: Split private keys across multiple parties
- **Threshold Signing**: Require minimum signatures for validity
- **Secure Reconstruction**: Combine shares without exposing full key
- **Byzantine Fault Tolerance**: Resilient to malicious participants

MPC Benefits

- **Enhanced Security**: No single point of key compromise
- **Distributed Trust**: Multiple parties required for signing
- **Fault Tolerance**: System continues with some parties offline
- **Regulatory Compliance**: Addresses custody and control requirements

Key Takeaways

Cryptography Implementation

- **Ed25519 Focus**: Solana's native cryptographic algorithm
- **Secure Key Management**: Private keys never exposed via API
- **Complete Transaction Flow**: From key generation to network submission
- **Audit Trail**: All operations recorded in database

SVM Cryptographic Advantages

- **High Performance**: Fast Ed25519 operations at scale
- **Parallel Processing**: Multiple signature verifications simultaneously
- **Network Security**: Cryptographic validation of all state changes
- **Future MPC Ready**: Foundation for advanced cryptographic schemes